The Frohmanian Tether on SDiff($\mathbb{T}^3$): A Metriplectic Correction, Its Canonicity, and an Obstruction to Damping

Conditional statements verified in Lean 4

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Companion repository: https://github.com/BenFrohman/frohmanian-tether-sdiff (version 0.1.2, draft)

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Abstract

We describe a Lean 4/Mathlib formalization of the *Frohmanian tether*, a symmetric quadratic correction $T(F,G) = -\kappa \|\omega\|_{L^2}^2 \int \langle \Pi_u \, \delta F/\delta\omega, \Pi_u \, \delta G/\delta\omega \rangle$ added to the Arnold Lie–Poisson bracket of ideal fluids, where Π_u is the L^2-orthogonal projection away from the velocity u and $\kappa = 3/2$. The library proves: the negative-feedback condition (C3) unconditionally; a uniqueness ("canonicity") statement inside a restricted kernel class; and an *obstruction to damping*: every symmetric kernel B satisfying the kinetic-energy degeneracy (C2) has $B(H,M) = B(M,H) = 0$ for every functional M, so such a kernel contributes no damping to any functional, in particular not to $\|\omega\|_{L^\infty}$. Neither (C1) nor (C2) is proved for T. Version 0.1.2 removes every hypothesis of earlier versions that turned out to be false or vacuous, with Lean refutations where possible: $\delta H/\delta\omega = u$, transitivity of the action on Casimir levels, the pointwise strain bound $\|\nabla u(x)\| \le C\|\omega\|_{L^\infty}$, the invariance hypothesis `TetherIntegralPreserved`, a circular Jacobi lemma, an antisymmetric uniqueness clause that can hold only for $T \equiv 0$, and a uniform lower bound on local existence times; it also repairs the definition of a physical solution, which in earlier versions admitted only solutions constant in time. On the analytic side the library proves a scalar comparison lemma for $y' \le Cy^2 - \kappa'' y^3$ and several conditional implications of the form "(cubic rate or uniform vorticity bound) + (Kato local existence) + (Beale–Kato–Majda continuation) + (uniqueness) $\Rightarrow$ global smooth solution", in which every analytic input is an explicit hypothesis. The rate and closing-bound hypotheses are open, are not supplied by the tether (its damping coefficient is 0), and are of the same strength as the global regularity problem. **This work does not prove global regularity of the three-dimensional Navier–Stokes equations and makes no claim toward the Clay Millennium Problem.** The build (Lean `v4.30.0-rc1`, Mathlib `v4.30.0-rc1`, 2715 jobs) contains 507 theorem constants (466 stated in the sources and 41 auto-generated equation, congruence and injectivity lemmas) and 750 declarations in 25 modules, with no `sorry`, no `native_decide`, no axiom declarations, and every declaration depending only on `propext`, `Classical.choice` and `Quot.sound`; this is enforced in continuous integration.

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1 Introduction and scope

Arnold's geometric picture of ideal hydrodynamics [1, 2] describes the Euler equations as Hamiltonian motion on coadjoint orbits of the group SDiff of volume-preserving diffeomorphisms, with Lie–Poisson bracket $\{F, G\}_{\mathrm{cl}}(\omega) = \int \omega \cdot (\delta F/\delta\omega \times \delta G/\delta\omega)$ and kinetic-energy Hamiltonian $H(\omega) = \frac{1}{2}\int |\mathrm{BS}\,\omega|^2$. The *Frohmanian tether* programme adds to this bracket a symmetric, negative-semidefinite quadratic term T and asks whether T can supply a damping mechanism for the vorticity sup-norm $M(t) = \|\omega(t)\|_{L^\infty}$, which by the Beale–Kato–Majda criterion [3] controls continuation of smooth Navier–Stokes solutions.

This paper reports exactly what the accompanying Lean 4 library proves. The Lean statements are the source of truth: every result below is stated to match a checked Lean declaration, with every hypothesis made explicit, and the exact Lean text is reproduced in Appendix A. The main findings are:

(i) The tether T satisfies (C3) unconditionally (Section 3). (C1) and (C2) are not proved. The identification $\delta H/\delta\omega = u$ assumed for (C2) in earlier versions is false with the library's definitions; this is proved in Lean (Theorem 3.3). The hypotheses formerly used for (C1) were false or unprovable and have been removed (Section 3.2).

(ii) *Obstruction to damping.* For every symmetric kernel B satisfying (C2), $B(H, M) = 0$ for every functional M; in particular, *if* T satisfied (C2), then $T(M, H) = T(H, M) = 0$ (Section 4). Either way, the symmetric design cannot both preserve the Euler vector field and produce the cubic damping rate on which the programme relied.

(iii) *Canonicity* holds only inside a restricted class and follows from saturation and polarization hypotheses; conditions (C1)–(C3) appear in the statement but are not used (Section 5).

(iv) The analytic layer consists of proved real-variable lemmas and *conditional* implications. Their remaining hypotheses are of two kinds (Section 9.1): classical theorems that are not in Mathlib (Kato local existence, Beale–Kato–Majda continuation, uniqueness), and hypotheses that are as strong as the open problem itself (a cubic rate or a uniform vorticity bound for all data) (Sections 7–8).

Scope statement. No result in this paper or in the library asserts global existence, smoothness or uniqueness of solutions of the three-dimensional Navier–Stokes equations. Every theorem whose conclusion mentions global solutions is an implication from hypotheses that are open. Section 9 lists precisely what the Lean development does and does not prove.

Version note. Version 0.1.0 of this manuscript named a different repository (`NS_Millennium_Proof`), quoted theorem and job counts of that repository, and cited a Zenodo DOI belonging to it. Version 0.1.1 refers only to the standalone repository above and uses the counts produced by its own build (Appendix B). Version 0.1.2 removes the hypothesis `KineticEnergyDerivIsVelocity` ($\delta H/\delta\omega = u$) and every theorem that assumed it or its pointwise form (the (C2) theorems `tetherKernel_degenerates_on_kinetic_energy`, `tetherKernel_C2_of_identifications` and `Combined.tetherKernel_C2_of_KEDeriv`; `tethered_reproduces_classical_euler`; `degeneracy_for_mollified_sup_norm_proxy`; the first clause of `frohmanian_tether_theorem`; the (C2) and no-damping conclusions of the combined capstone), together with the lemmas `functional_derivative_eq_velocity_of_unique_repr` and its two aliases, whose hypotheses can never hold (Theorem 3.4). It adds the Lean refutation of the hypothesis (Section 3.1). It further removes the false or vacuous hypotheses listed in Section 3.2, together with every theorem that depended on them, and repairs `IsPhysicalNSSolution` (Section 2). No DOI is assigned to this version in the manuscript.

2 The formal setting

We record the definitions as formalized, including the points at which they differ from the classical objects.

- **Space.** `T3` is `EuclideanSpace ℝ (Fin 3)`, i.e. $\mathbb{R}^3$ with Lebesgue measure, *not* the quotient torus $\mathbb{R}^3/\mathbb{Z}^3$. Periodicity, where needed (Section 8), is the predicate `IsZ3Periodic` on functions on $\mathbb{R}^3$. The notation SDiff($\mathbb{T}^3$) in the title refers to the motivating geometry; no diffeomorphism group of $\mathbb{T}^3$ is constructed.

- **Phase space.** `CoadjointOrbit` is the type of pointwise divergence-free vector fields $\omega : \mathbb{R}^3 \to \mathbb{R}^3$ (a subtype; no orbit structure is built in).

- **Functional derivative.** `FunctionalDerivative F ω` is a *choice* of a Gâteaux representative δ of F at ω along C^1 divergence-free variations, and is 0 if none exists. A Gâteaux representative is never unique (Theorem 3.4: changing it at a single point gives another one), and at a non-differentiable ω every field is a representative, so the chosen field is not canonical.

- **Velocity and energy.** `velocity_from_vorticity ω` is the Biot–Savart integral `BiotSavart ω.val`, and `KineticEnergyHamiltonian ω` $= \frac{1}{2}\int \|\mathrm{BS}\,\omega\|^2$.

- **Navier–Stokes.** `navier_stokes_eq u p ν` (and `NS_PDE`) asserts the momentum equation with Mathlib's total derivative `deriv` in time, and $\nabla \cdot u = 0$, pointwise for $t \geq 0$. `IsPhysicalNSSolution u` adds pointwise time-differentiability for $t \geq 0$, spatial continuity, finite kinetic energy, the energy inequality $\int \|u(t)\|^2 \leq \int \|u(0)\|^2$, and an *affine extension to negative times*: there is a field v_0 with $u(t) = u(0) + t\,v_0$ for all $t \leq 0$.

Repair of `IsPhysicalNSSolution`. Earlier versions required $u(t) = u(0)$ for $t \leq 0$. Since `deriv` is the two-sided derivative, differentiability at $t = 0$ then forced $\partial_t u(0) = 0$, so a solution whose velocity changes at the initial time could never qualify; with the energy inequality, this made the class essentially one of solutions that are stationary at $t = 0$. The

negative-time clause exists only to stop junk values at $t < 0$ from interacting with `deriv` at 0; the affine extension does this and allows any initial time derivative v_0. The sanity lemma `PhysicalSolutionSanity.exists_physical_time_deriv_ne_zero` exhibits a path u satisfying `IsPhysicalNSSolution` whose time derivative at $t = 0$ is nonzero at some point (`not_forall_physical_time_deriv_zero` states the same as the failure of the old consequence). The path is $g(t)\,\varphi$ with $\varphi(x) = \max(0, 1 - \|x\|)\,e_0$, $g(t) = 1 - t$ for $t \le 0$ and $g(t) = 1/(1+t)$ for $t > 0$; it is *not* a Navier–Stokes solution. The library contains no nonzero Navier–Stokes solution, so non-vacuity of the class for genuine non-steady flows is not formalized. The exact definitions are quoted in Appendix A.1.

3 The tether kernel and conditions (C1)–(C3)

Definition 3.1 (`Pi_u`, `TetherKernel`, `TetheredBracket`)**.** For velocity fields u, v put $\Pi_u v = v - \big(\int \langle v, u\rangle / \int \|u\|^2\big)\,u$ (in Lean, division by 0 is 0, so $\Pi_u v = v$ when $\int \|u\|^2 = 0$). With $\kappa = 3/2$ (`kappa_eq_three_halves`), $\mathcal{E}(\omega) = \int \|\omega\|^2$ (`Enstrophy`) and $u = \mathrm{BS}\,\omega$,

$$T(\omega; F, G) = -\kappa\,\mathcal{E}(\omega) \int \big\langle \Pi_u\,\delta F/\delta\omega,\ \Pi_u\,\delta G/\delta\omega \big\rangle\,dx, \qquad \{F, G\}_{\mathfrak{T}} = \{F, G\}_{\mathrm{cl}} + T(F, G).$$

The three conditions are predicates on a kernel $B(\omega; F, G)$: (C1) `InvariantUnderCoadjointAction`: $B(g\cdot\omega; F, G) = B(\omega; F, G)$ for all g in a group `SDiff` acting on `CoadjointOrbit`; (C2) `DegenerateWRTKineticEnergy`: $B(\omega; F, H) = 0$ for all F, ω; (C3) `ProducesControllableNegativeFeedback`: $B(\omega; F, F) \le -\kappa\,\mathcal{E}(\omega) \int \|\Pi_u\,\delta F/\delta\omega\|^2$.

Theorem 3.2 (Symmetry and (C3); `tetherKernel_comm`, `tetherKernel_C3`)**.** *For all ω, F, G: $T(\omega; F, G) = T(\omega; G, F)$, and T satisfies (C3). No hypotheses.*

(C2) is not proved. No theorem of the library asserts that T satisfies (C2). The only (C2) fact proved for T is the degenerate case `tetherKernel_C2_of_no_gateaux`: if H has no Gâteaux representative at ω, then $T(\omega; F, H) = 0$ for every F (because the encoding sets the derivative to 0). Section 3.1 explains why the former route to (C2) was removed.

(C1) is not proved. Earlier versions derived (C1) for T from two hypotheses, `SDiffActionHyp` with a transitivity field and `TetherIntegralPreserved`; both are removed (Section 3.2) and no (C1) theorem for T remains. `SDiffActionHyp` now has the single field `casimir_invariant` (every Casimir $\int f(|\omega|)$ is invariant under the action). For the concrete group `ConcreteSDiff.SDiffConcrete` of measure-preserving, measurable, divergence-preserving permutations of $\mathbb{R}^3$ acting by $g \cdot \omega = \omega \circ g^{-1}$ (scalar composition, not the geometric push-forward $(Dg\,\omega)\circ g^{-1}$), the structure is proved outright (`ConcreteSDiff.sdiffActionHyp`).

3.1 The identification $\delta H/\delta\omega = u$ is false (removed)

Earlier versions used the named proposition `PlanSteps.KineticEnergyDerivIsVelocity`: *for every ω, `FunctionalDerivative KineticEnergyHamiltonian ω = velocity_from_vorticity ω`*, as the hypothesis of (C2) for T. It has been deleted from the library, together with every theorem that assumed it or its pointwise form, because it is false.

Theorem 3.3 (Refutation of $\delta H/\delta\omega = u$)**.** *(Lean: `KineticEnergyDerivative.not_forall_functionalDerivative_kineticEnergy_eq_velocity`.) It is not the case that `FunctionalDerivative KineticEnergyHamiltonian ω = velocity_from_vorticity ω` for every `ω : CoadjointOrbit`.*

Proof idea (formalized). Let B be the open unit ball centred at $(0,2,0)$ and $\omega = \mathbf{1}_B\, e_1$. Each coordinate of ω and of 2ω takes two values, so its Fréchet derivative is 0 everywhere (Mathlib's `fderiv` is 0 at non-differentiable points, and a two-valued function differentiable at a point is locally constant there); hence $\nabla \cdot \omega = 0$ in the library's sense and $\omega, 2\omega$ are points of `CoadjointOrbit`. Neither is differentiable (the first coordinate jumps at $(1,2,0)$), so the defining condition of a Gâteaux representative is vacuous and `FunctionalDerivative` returns the same choice value at both. If the hypothesis held, $\mathrm{BS}(2\omega) = \mathrm{BS}\,\omega$; but $\mathrm{BS}(2\omega) = 2\,\mathrm{BS}\,\omega$ and the third component of $\mathrm{BS}\,\omega$ at the origin is $-(4\pi)^{-1}\int_B y_2\,|y|^{-3}\,dy < 0$. Theorem 3.3 uses only the three standard axioms.

Theorem 3.4 (Non-uniqueness of Gâteaux representatives)**.** *(Lean: `KineticEnergyDerivative.not_unique_isGateauxRepresentative`, `KineticEnergyDerivative.not_unique_velocity_representative`.) If δ is a Gâteaux representative of F at ω, so is a field $\delta' \neq \delta$ (change δ at the origin). Consequently, for no ω is* $\mathrm{BS}\,\omega$ *the* unique *Gâteaux representative of H at ω.*

Scope of the refutation. The counterexample is a non-smooth vorticity field, at which the encoding's derivative is an unconstrained choice; it shows that the Lean hypothesis, as stated, is false, and therefore that every theorem which assumed it was vacuous. It does not by itself settle what the derivative is at smooth fields. For the vorticity pairing used by `IsGateauxRepresentative`, the classical formal computation gives $\delta H/\delta\omega = \mathrm{BS}^*\mathrm{BS}\,\omega$, the vector stream function ψ with $\nabla \times \psi = u$, not u; this identity is *not* formalized. What is formalized is the velocity-pairing statement `isVelocityGateauxRepresentative_kineticEnergy_at`: with the pairing $\int \langle \delta, \mathrm{BS}\,\varepsilon\rangle$, the derivative of H along ε is represented by u (under explicit integrability hypotheses). Whether T, with ψ in place of u, satisfies (C2) is open and in general expected to fail: $\Pi_u \psi = 0$ would require ψ to be parallel to u.

3.2 Other removed hypotheses (version 0.1.2)

Each item below was an explicit hypothesis of earlier versions. Each was false, unprovable in the encoding, or vacuous, and has been deleted together with every theorem that used it. No remaining statement was weakened silently: a theorem whose proof needed a removed hypothesis was deleted, not re-proved under a weaker one. All refutations use only the three standard axioms.

1. **Transitivity on Casimir levels** (field `transitive_on_casimir_levels` of `SDiffActionHyp`: fields with equal Casimirs lie in one orbit). *Refuted* for the concrete action: `ConcreteSDiff.not_transitive_on_casimir_levels`. The constant fields e_0 and e_1 have the same Casimirs but different orbits, because the action fixes every constant field. Deleted with it: `orbit_eq_casimir_level`.

2. **`TetherIntegralPreserved`** (invariance of $\int \|\Pi_u\, \delta F/\delta\omega\|^2$ under the action). It concerns the unconstrained choice made by `FunctionalDerivative`, so it can be neither proved nor refuted from the definitions (see the paragraph on degeneracy below). Deleted with it: `tether_coadjoint_invariance`, `tether_satisfies_C1`, `tetherKernel_C1`, `Combined.tetherKernel_C1_of_casimir`, `Combined.tetherKernel_C1_of_volumePreserving` (whose structure field carried the same assumption), `Combined.combined_conditional_capstone`, and related bridges.

3. **Pointwise strain bound** `PointwiseStrainSupBound`: $\|\nabla u(x)\| \le C\|\omega\|_{L^\infty}$. *Refuted*: `StrainBoundRefutation.not_forall_strain_le_sup_vorticity`. For every C, the linear strain field $u(y) = (y_1, y_0, 0)$ has $\nabla \times u = 0$ and $\nabla u \neq 0$. (Classically the bound fails even for smooth decaying fields, since the Calderón–Zygmund operator is

unbounded on L^∞.) Deleted with it: the stretching estimates at a spatial maximum (`stretching_inner_le_at_max` and relatives), `tethered_stretching_of_weight`, `reduced_pairing_of_unmodified_and_tether` and the other conditional bridges built on them.

4. **Circular Jacobi lemma.** `tethered_jacobi_identity` assumed that the Jacobiator equals a cyclic integral (`hident`) and that this integral vanishes (`hvanish`), i.e. its conclusion. Deleted. No Jacobi identity is claimed for $\{\cdot,\cdot\}_{\mathfrak{T}}$.

5. **Antisymmetric uniqueness clause.** The former first clause of `frohmanian_tether_theorem` (and `uniqueness_of_minimal_tether_antisymmetric`) quantified over antisymmetric kernels that saturate T. Proved: `antisymmetric_clause_forces_zero`. Antisymmetry, saturation and the two polarization hypotheses force $B = 0$ *and* $T = 0$, so the clause applied to no kernel unless $T \equiv 0$. Deleted.

6. **Uniform lower bound `hTlb` on local existence times**, with the restart hypotheses `hInt`, `huniq_shift` of `frohmanian_tether_theorem`. A data-independent lower bound on Kato existence times for all smooth finite-energy data is not known. By Navier–Stokes scaling it is of the same strength as the open problem; it is not known to be false. It was removed and the theorem re-proved through the BKM route (Theorem 7.5).

Degeneracy of the encoding. `functionalDerivative_indep_of_not_differentiable`: at a vorticity with a non-differentiable component, `FunctionalDerivative F ω` does not depend on F. Consequently `tetherKernel_indep_of_not_differentiable` holds, and `polarizes_tetherKernel_zero_of_not_differentiable` shows that the hypothesis `Polarizes TetherKernel` forces $T(\omega; F, G) = 0$ at every such ω. Because `FunctionalDerivative` is an unspecified choice among non-unique representatives, the kernel-class conditions of the canonicity theorems (`Polarizes TetherKernel`, `h_proj`, `h_nondeg`: "T is not identically zero") cannot be proved from the definitions. They remain explicit hypotheses (Section 9.1).

4 The obstruction to damping

Theorem 4.1 (`Combined.left_H_zero_of_C2_symm`)**.** *Let B be any kernel that is symmetric, $B(\omega; F, G) = B(\omega; G, F)$, and satisfies (C2). Then $B(\omega; H, M) = 0$ for every ω and every functional M.*

Corollary 4.2 (`Combined.tetherKernel_left_H_zero`)**.** *If T satisfies (C2), then $T(\omega; H, M) = T(\omega; M, H) = 0$ for every ω and M.*

Theorem 4.1 is unconditional. For the tether, Corollary 4.2 requires (C2) as a hypothesis, and (C2) for T is not proved (Section 3.1). The interpretation, for T or any symmetric kernel that satisfies (C2), is the following. Along the tethered flow $\dot{M} = \{M, H\}_{\mathfrak{T}} = \{M, H\}_{\mathrm{cl}} + T(M, H)$, the tether term would vanish; hence the symmetric design that would make the tether preserve the Euler vector field (C2) would also force its contribution to the evolution of every functional, including smoothed versions of $\|\omega\|_{L^\infty}$, to be zero. The negative diagonal values $T(F, F) \le 0$ of (C3) never enter dynamics generated by H. Consequently the cubic rate $M' \le CM^2 - \kappa'' M^3$ used below is not a consequence of the tether: under (C2) its damping coefficient would be 0, and without (C2) nothing about it is proved.

A related formal fact: `PlanSteps.tether_bound_not_scale_invariant`: for $\kappa > 0$, $M \neq 0$ and any real S, the family of inequalities $\lambda^6 S \le C(\lambda^2 M)^3 - \kappa(\lambda^2 M)^4$ $(\lambda > 0)$ fails for some λ, so a pointwise bound of this shape cannot be invariant under Navier–Stokes scaling.

5 Canonicity

Theorem 5.1 (`uniqueness_of_minimal_tether`; delegates to `uniqueness_of_minimal_symmetric_tether`)**.** *Let a group act on* `CoadjointOrbit` *and assume* `SDiffActionHyp` *(proved for the concrete group). Let B be symmetric, satisfy (C1), (C2), (C3),* `SaturatesTetherQuadratic B` *(i.e. $B(\omega; F, F) = T(\omega; F, F)$ for all F, ω),* `Polarizes B` *and* `Polarizes TetherKernel` *(the polarization identity $2B(F, G) = B(F+G, F+G) - B(F, F) - B(G, G)$). Then $B = T$.*

Remark 5.2. The proof uses only saturation and the two polarization hypotheses: a kernel that agrees with T on the diagonal and satisfies polarization equals T. The hypotheses `SDiffActionHyp` and (C1)–(C3) are not used. Theorem 5.1 is therefore a uniqueness statement *given* agreement on the diagonal, not a derivation of T from (C1)–(C3). The hypothesis `Polarizes TetherKernel` is not proved and forces T to vanish at every non-differentiable vorticity (`polarizes_tetherKernel_zero_of_not_differentiable`).

Theorem 5.3 (`uniqueness_of_minimal_tether_core`)**.** *Let B be symmetric and assume: $B(\omega; F, G)$ depends on F only through $\Pi_u \, \delta F/\delta\omega$ (`h_proj`); if B factors through the projection then it has a density α (`HasTetherKernelDensity`) (`h_dens`); every such density is of the form $\alpha(\omega, x) = -c\,\mathcal{E}(\omega)$ (`h_weight`); B saturates T on the diagonal (`h_sat`); and there are ω, F with $\mathcal{E}(\omega) \int \langle \Pi_u \delta F/\delta\omega, \Pi_u \delta F/\delta\omega \rangle \neq 0$ (`h_nondeg`). Then $B = T$.*

6 Bracket structure and Casimirs

- `classicalBracket_antisymm`: $\{F, G\}_{\mathrm{cl}} = -\{G, F\}_{\mathrm{cl}}$; `classicalBracket_H_H`: $\{H, H\}_{\mathrm{cl}} = 0$. The Jacobi identity of $\{\cdot, \cdot\}_{\mathrm{cl}}$ is classical [1] and is *not* formalized.
- Since T is symmetric and in general nonzero, $\{\cdot, \cdot\}_{\mathfrak{T}}$ is *not* antisymmetric; it is of metriplectic type in the sense of Morrison [7] (antisymmetric part plus symmetric negative-semidefinite part). No Jacobi identity is claimed for it.
- `PlanSteps.casimir_conserved`: under `SDiffActionHyp` and `IsCoadjointMotion` (the trajectory is $\omega(t) = g(t) \cdot \omega(0)$), every Casimir is constant along the trajectory. That the Euler flow is such a motion is Arnold's classical picture but is not formalized (Section 9.1).
- `ScalingIdentities.vorticity_rescale`: for $u_\lambda(x) = \lambda u(\lambda x)$, $\nabla \times u_\lambda(x) = \lambda^2 (\nabla \times u)(\lambda x)$ under differentiability at λx.

7 The analytic layer

Lemma 7.1 (Comparison; `AnalyticPipeline.comparison_ode_stability`)**.** *Let $C, \kappa'' > 0$ and let $y : \mathbb{R} \to \mathbb{R}$ be differentiable at every $s \geq 0$ with $y'(s) \leq C y(s)^2 - \kappa'' y(s)^3$ for $s \geq 0$. Then $y(t) \leq \max(y(0), C/\kappa'')$ for every $t \geq 0$.*

Proposition 7.2 (`riccati_ceiling_of_vorticity_di`)**.** *If $t \mapsto M(t) = \sup_x \|(\nabla \times u(t))(x)\|$ is continuous, differentiable at every $s \geq 0$, and satisfies $M'(s) \leq C M(s)^2 - \kappa'' M(s)^3$ for $s \geq 0$ with $C, \kappa'' > 0$, then M is bounded on $[0, \infty)$.*

The following hypotheses are named propositions (exact text in Appendix A.4); all are open for general data:

- `FrohmanianTetherRate u₀ ν` (R): for $\nu > 0$ and every physical solution from u_0, M is continuous, differentiable on $[0, \infty)$, pointwise bounded in space, and satisfies the cubic inequality for some $C, \kappa'' > 0$. (Differentiability of a sup-norm in time is itself a strong requirement.)

- `FrohmanianTetherBound u₀ ν`: every physical solution from u_0 has $\sup_{t\geq 0} M(t) < \infty$ (with continuity and spatial boundedness). `ClayStatements.FrohmanianTetherClosingBound ν` asserts this for *all* smooth divergence-free finite-energy data.
- `KatoLocalWitness u₀ ν`: data of a solution, smooth on $[0, T]$ with $T > 0$, that is a physical solution (local existence [6] is not formalized; a witness is an input). *Caveat:* the field `pde : NS_PDE u p ν` asks for the equations at *every* $t \geq 0$, not only on $[0, T]$, so as encoded a witness is stronger than the conclusion of Kato's local theorem.
- `PhysicalBKMContinuationCriterionFor u₀ ν`: the Beale–Kato–Majda criterion [3] for physical solutions from u_0, stated via the maximal time `Tstar`; not formalized, an input.
- `KatoUniquenessOnExistenceTimes u₀ ν w`: every smooth physical solution from u_0 agrees with the witness w on each existence interval; an input.

Theorem 7.3 (`ClosureBlueprint.conditional_regularity_of_open_hyps`)**.** *Let $\nu > 0$ and u_0 smooth, divergence-free, with $\|u_0\|^2$ integrable. Assume `KatoLocalWitness u₀ ν`, `FrohmanianTetherBound u₀ ν`, `PhysicalBKMContinuationCriterionFor u₀ ν` and `KatoUniquenessOnExistenceTimes u₀ ν w`. Then there exist u, p with `NS_PDE u p ν`, $u(0) = u_0$, $u(t)$ smooth and divergence-free for every $t \geq 0$ (and $M(t) \geq 0$).*

Theorem 7.4 (`ConditionalRegularity.conditional_regularity_of_open_hyps`)**.** *Under the hypotheses of Theorem 7.3 (stated for a time-dependent field $u_0(\cdot)$ that is divergence-free at all times, with $u_0(0)$ smooth and of finite energy), there is a u with $u(0) = u_0(0)$, $u(t)$ smooth and satisfying `satisfies_NavierStokes` (physical solution of the equations) for $t \geq 0$, and unique among smooth fields with these properties.*

Theorem 7.5 (`frohmanian_tether_theorem`)**.** *Assume, for all smooth divergence-free finite-energy u_0 and $\nu > 0$: (hKato) a Kato local witness; (hDI) `FrohmanianTetherRate u₀ ν`; (hcrit) `PhysicalBKMContinuationCriterionFor u₀ ν`; and (huniq) `KatoUniquenessOnExistenceTimes u₀ ν w` for every witness w. Then there is a kernel $\mathfrak{T}_F$ such that: (1) every kernel with density equal to the canonical tether density equals $\mathfrak{T}_F$; and (2) for every $\nu > 0$ and every smooth divergence-free finite-energy u_0 there are u, p with `NS_PDE u p ν`, $u(0) = u_0$, and $u(t)$ smooth and divergence-free for all $t \geq 0$.*

Remark 7.6. Clause (2) is a global-existence statement *conditional on* (hDI) for all data, which is as strong as the open problem, together with the classical inputs (hKato), (hcrit), (huniq); it is not a regularity theorem. The proof goes through the comparison lemma (bounded M), then BKM continuation and uniqueness. Version 0.1.1 had a clause assuming $\delta H/\delta\omega = u$; draft versions also had an antisymmetric clause and the hypotheses `hTlb`, `hInt`, `huniq_shift`. All were removed (Section 3.2).

Theorem 7.7 (Non-vacuity at zero data; `conditional_regularity_nonvacuous_at_zero`)**.** *For every $\nu > 0$ there is u with $u(0) = 0$, smooth and a physical solution for $t \geq 0$, unique among smooth physical solutions with zero data. Also `stepField_not_physical`: a time-discontinuous step field, which would satisfy `NS_PDE` because `deriv` is 0 at non-differentiable points, is excluded by `IsPhysicalNSSolution`.*

This shows only that the conclusion form of Theorem 7.4 is consistent in the trivial case; it gives no information on nonzero data.

8 The periodic Clay statement and a conditional reduction

`ClayStatements.Clay_B_exact` formalizes a version of Fefferman's statement (B) [4]: for every $\nu > 0$ and every smooth, divergence-free, $\mathbb{Z}^3$-periodic u_0 on $\mathbb{R}^3$ there is a solution (u, p) that

is C^∞ on $[0,\infty)\times\mathbb{R}^3$, solves the equations with zero forcing, has $u(0)=u_0$, and is periodic for all $t\geq 0$. (Fefferman's formulation also contains a bounded-energy condition; the formal statement is as quoted in Appendix A.5.)

Theorem 8.1 (`clay_B_exact_of_periodic_tether_bound`)**.** *If* `PeriodicTetherClosingBound ν` *and* `PeriodicBKMContinuation ν` *hold for every* $\nu>0$*, then* `Clay_B_exact`*.*

`PeriodicTetherClosingBound` is a uniform vorticity bound for all periodic smooth data and `PeriodicBKMContinuation` says that such a bound yields a global periodic smooth solution. Together they are at least as strong as the conclusion; the reduction is a formal bookkeeping step and makes no progress on (B).

9 What the Lean development does and does not prove

Proved (no further hypotheses). Symmetry of T and (C3); antisymmetry of $\{\cdot,\cdot\}_{\mathrm{cl}}$ and $\{H,H\}_{\mathrm{cl}}=0$; Theorem 4.1 ($B(H,M)=0$ for every symmetric B satisfying (C2)); the refutation Theorem 3.3 of $\delta H/\delta\omega=u$ and the non-uniqueness Theorem 3.4; `SDiffActionHyp` for the concrete permutation group; the refutations of Section 3.2 and the degeneracy lemmas of the encoding; the sanity lemma for `IsPhysicalNSSolution`; the comparison Lemma 7.1; the scaling identity and the scale non-invariance lemma; the zero-data facts and `stepField_not_physical`; the Fourier-symbol identities `PlanSteps.bsSymbol_div`, `PlanSteps.bsSymbol_curl` (mode-wise algebra, not connected to the real-space Biot–Savart integral).

Proved as implications from named, open hypotheses. Canonicity (from saturation and polarization); Casimir conservation (from `IsCoadjointMotion`); the conditional regularity theorems 7.3, 7.4, 7.5 (from rate or closing-bound hypotheses, Kato existence, BKM continuation and uniqueness); the Clay (B) reduction. The hypotheses are listed by category in Section 9.1.

Not proved, and not claimed.

- Global existence, smoothness or uniqueness for 3D Navier–Stokes with large data; any part of the Clay Millennium Problem.
- The cubic rate `FrohmanianTetherRate`, any closing bound, or any damping mechanism from the tether (under (C2) its damping coefficient would be 0).
- (C2) for T. The identification $\delta H/\delta\omega=u$ is refuted (Theorem 3.3); the classical identification $\delta H/\delta\omega=\psi$ (stream function) is not formalized.
- (C1) for T. That the Euler flow is a coadjoint motion.
- A Jacobi identity for $\{\cdot,\cdot\}_{\mathfrak{T}}$ (the former conditional lemma was circular and is deleted), or a Poisson structure; the Jacobi identity for $\{\cdot,\cdot\}_{\mathrm{cl}}$ (cited only).
- Kato local existence [6], Beale–Kato–Majda [3], parabolic smoothing, energy uniqueness: all are inputs, none is formalized.
- Anything on the quotient torus: `T3` is $\mathbb{R}^3$, and no diffeomorphism group of $\mathbb{T}^3$ is constructed.
- Derivation of T from (C1)–(C3): canonicity assumes diagonal agreement with T.
- Any nonzero Navier–Stokes solution; the sanity path for `IsPhysicalNSSolution` is not one.

The physical motivations of earlier versions (fluid–gravity duality, Onsager's conjecture [5], metriplectic viscous completion) are not formalized and play no role in any statement here.

9.1 Remaining assumptions, by category

Every remaining assumption is an explicit Lean argument or a named `Prop`. None is an axiom.

(b) Classical results not in Mathlib.
`KatoLocalWitness` (Kato local existence [6]; as encoded it is stronger, see Section 7); `PhysicalBKMContinuationCriterionFor` and `PeriodicBKMContinuation` (Beale–Kato–Majda continuation [3]); `KatoUniquenessOnExistenceTimes` (uniqueness of smooth solutions); `IsCoadjointMotion` for the Euler flow (Arnold [1]).

(d) Equivalent in strength to the open problem.
`FrohmanianTetherRate` (the cubic rate for every physical solution; it also asks for time-differentiability of $\|\omega(t)\|_{L^\infty}$, which can fail for genuine flows); `FrohmanianTetherBound` and `ClayStatements.FrohmanianTetherClosingBound` (uniform vorticity bounds); `ClayStatements.PeriodicTetherClosingBound`. The tether does not supply any of these.

(c) Kernel-class conditions, not provable in the current encoding.
In the canonicity theorems: `SaturatesTetherQuadratic B`, `Polarizes B`, `Polarizes TetherKernel`, `h_proj`, `h_dens`, `h_weight`, `h_nondeg` (T not identically zero). (C2) for T, assumed in Corollary 4.2. These are hypotheses on B or on the choice made by `FunctionalDerivative` (Section 3.2), not analytic facts about Navier–Stokes.

A Exact Lean statements

Quoted verbatim from the repository by `scripts/extract_statements.py`; proof bodies are replaced by `....` Each block starts with its file and line.

A.1 Setting

```
-- FrohmanianTether/ArnoldGeometric.lean, line 172
public structure CoadjointOrbit where
  val : T3 → EuclideanSpace ℝ (Fin 3)
  property : ∀ x, NavierStokes3D.div val x = 0
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 308
public noncomputable def FunctionalDerivative (F : CoadjointOrbit → ℝ)
    (ω : CoadjointOrbit) : T3 → EuclideanSpace ℝ (Fin 3) :=
  if h : ∃ δ, IsGateauxRepresentative F ω δ then Classical.choose h else 0
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 947
public noncomputable def KineticEnergyHamiltonian (ω : CoadjointOrbit) : ℝ :=
  (1/2) * ∫ x, ‖BiotSavart ω.val x‖^2 ∂(volume)
```

```
-- FrohmanianTether/NS_Equations.lean, line 2149
@[expose] public def navier_stokes_eq (u : TimeDependentVelocity) (p : TimeDependentPressure)
    (ν : ℝ) : Prop :=
  ∀ t ≥ (0 : ℝ), ∀ x : T3,
    time_deriv u t x + convective (u t) (u t) x + pressureGradient (p t) x =
        ν • laplacian (u t) x ∧
      div (u t) x = 0
```

```
-- FrohmanianTether/NS_Equations.lean, line 2270
@[expose] public def IsPhysicalNSSolution (u : TimeDependentVelocity) : Prop :=
  (∀ t ≥ (0 : ℝ), ∀ x : T3, DifferentiableAt ℝ (fun s => u s x) t) ∧
  (∀ t ≥ (0 : ℝ), Continuous (u t)) ∧
  (∀ t ≥ (0 : ℝ), Integrable (fun x : T3 => ‖u t x‖ ^ 2)) ∧
  (∀ t ≥ (0 : ℝ), (∫ x, ‖u t x‖ ^ 2 ∂volume) ≤ ∫ x, ‖u 0 x‖ ^ 2 ∂volume) ∧
  (∃ v₀ : VelocityField, ∀ t ≤ (0 : ℝ), u t = u 0 + t • v₀)
```

```
-- FrohmanianTether/PhysicalSolutionSanity.lean, line 144
public theorem exists_physical_time_deriv_ne_zero :
    ∃ u : TimeDependentVelocity, IsPhysicalNSSolution u ∧
      ∃ x : T3, deriv (fun s => u s x) 0 ≠ 0 := ...
```

```
-- FrohmanianTether/PhysicalSolutionSanity.lean, line 157
public theorem not_forall_physical_time_deriv_zero :
    ¬ ∀ u : TimeDependentVelocity, IsPhysicalNSSolution u →
      ∀ x : T3, deriv (fun s => u s x) 0 = 0 := ...
```

```
-- FrohmanianTether/NS_Equations.lean, line 2433
public structure KatoLocalWitness (u₀ : VelocityField) (ν : ℝ) where
  T : ℝ
  Tpos : 0 < T
  u : TimeDependentVelocity
  p : TimeDependentPressure
  smooth : ∀ t ∈ Set.Icc (0 : ℝ) T, ContDiff ℝ (⊤ : ℕ∞) (u t)
  init : u 0 = u₀
  pde : NS_PDE u p ν
  phys : IsPhysicalNSSolution u
```

A.2 Tether, (C1)–(C3), obstruction, canonicity

```
-- FrohmanianTether/ForMathlib/Projection.lean, line 35
@[expose] public noncomputable def Pi_u (u v : VelocityField) : VelocityField :=
  fun x => v x - (
    (∫ y, pairing (v y) (u y) ∂volume) /
    (∫ y, ‖u y‖ ^ 2 ∂volume)
  ) • u x
```

```
-- FrohmanianTether/SymplecticTether.lean, line 351
@[expose] public noncomputable def TetherKernel (ω : CoadjointOrbit) (F G : Functional) : ℝ :=
  -kappa * Enstrophy ω * ∫ x,
      inner ℝ
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x)
    ∂volume
```

```
-- FrohmanianTether/SymplecticTether.lean, line 358
public noncomputable def TetheredBracket (F G : Functional) (ω : CoadjointOrbit) : ℝ :=
  -- The Frohmanian Symplectic Tether bracket: the classical Arnold Lie–Poisson bracket
  -- plus the minimal quadratic correction (the tether kernel) that satisfies (C1)–(C3).
  -- Canonicity is proved only inside the restricted kernel class and under the explicit
  -- hypotheses of `uniqueness_of_minimal_tether` (Uniqueness.lean); it is NOT an
  -- unconditional uniqueness statement. The correction is symmetric, so this bracket is
  -- not antisymmetric.
  ClassicalBracket F G ω + TetherKernel ω F G
```

```
-- FrohmanianTether/SymplecticTether.lean, line 375
@[expose] public def InvariantUnderCoadjointAction (SDiff : Type*) [Group SDiff]
    [MulAction SDiff CoadjointOrbit] (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop
    ↪  :=
  -- (C1): The correction B is invariant under the coadjoint action of SDiff(T³).
  -- For all volume-preserving g, the value of the correction is unchanged when the vorticity
  -- and the test functionals are transformed by the coadjoint action.
  ∀ (g : SDiff) (F G : Functional) (ω : CoadjointOrbit),
    B (CoadjointAction g ω) F G = B ω F G
```

```
-- FrohmanianTether/SymplecticTether.lean, line 383
@[expose] public def DegenerateWRTKineticEnergy (B : CoadjointOrbit → Functional → Functional
↪  → ℝ) : Prop :=
  -- (C2): The correction is degenerate with respect to the kinetic-energy Hamiltonian H.
  -- The Hamiltonian vector field generated by H is exactly the classical one (no modification
  -- to the reversible Euler dynamics). This is the energy-preserving degeneracy on energy.
  ∀ (F : Functional) (ω : CoadjointOrbit),
    B ω F KineticEnergyHamiltonian = 0
```

```
-- FrohmanianTether/SymplecticTether.lean, line 391
@[expose] public def ProducesControllableNegativeFeedback
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (F : Functional) (ω : CoadjointOrbit),
    B ω F F ≤
      -kappa * Enstrophy ω * ∫ x,
          ‖Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x‖ ^ 2
        ∂volume
```

```
-- FrohmanianTether/SymplecticTether.lean, line 414
public theorem tetherKernel_comm (ω : CoadjointOrbit) (F G : Functional) :
    TetherKernel ω F G = TetherKernel ω G F := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 445
public theorem tetherKernel_C3 (F : Functional) (ω : CoadjointOrbit) :
    TetherKernel ω F F ≤
      -kappa * Enstrophy ω * ∫ x,
          ‖Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x‖ ^ 2
        ∂volume := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 792
public theorem tetherKernel_C2_of_no_gateaux
    (ω : CoadjointOrbit)
    (h : ¬ ∃ dH, IsGateauxRepresentative KineticEnergyHamiltonian ω dH)
    (F : Functional) :
    TetherKernel ω F KineticEnergyHamiltonian = 0 := ...
```

```
-- FrohmanianTether/KineticEnergyDerivative.lean, line 295
theorem not_forall_functionalDerivative_kineticEnergy_eq_velocity :
    ¬ ∀ ω : CoadjointOrbit,
      FunctionalDerivative KineticEnergyHamiltonian ω = velocity_from_vorticity ω := ...
```

```
-- FrohmanianTether/KineticEnergyDerivative.lean, line 324
theorem not_unique_isGateauxRepresentative (F : CoadjointOrbit → ℝ) (ω : CoadjointOrbit)
    (d : T3 → EuclideanSpace ℝ (Fin 3)) (hd : IsGateauxRepresentative F ω d) :
    ∃ d', IsGateauxRepresentative F ω d' ∧ d' ≠ d := ...
```

```
-- FrohmanianTether/KineticEnergyDerivative.lean, line 349
theorem not_unique_velocity_representative (ω : CoadjointOrbit) :
    ¬ (IsGateauxRepresentative KineticEnergyHamiltonian ω (velocity_from_vorticity ω) ∧
      ∀ dH, IsGateauxRepresentative KineticEnergyHamiltonian ω dH →
        dH = velocity_from_vorticity ω) := ...
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 1035
public theorem isVelocityGateauxRepresentative_kineticEnergy_at
    (ω ε : CoadjointOrbit)
    (hω : ∀ i x, DifferentiableAt ℝ (fun y => ω.val y i) x)
    (hε : ∀ i x, DifferentiableAt ℝ (fun y => ε.val y i) x)
    (hkerω : ∀ x, Integrable
      (fun y => biotSavartKernel x y • cross (ω.val y) (x - y)))
    (hkerε : ∀ x, Integrable
      (fun y => biotSavartKernel x y • cross (ε.val y) (x - y)))
    (hF_meas : ∀ᶠ t in nhds (0 : ℝ),
      AEStronglyMeasurable
        (fun x => (1 / 2 : ℝ) *
          ‖BiotSavart ω.val x + t • BiotSavart ε.val x‖ ^ 2)
        NavierStokes3D.volume)
    (hF_int : Integrable
      (fun x => (1 / 2 : ℝ) * ‖BiotSavart ω.val x‖ ^ 2)
      NavierStokes3D.volume)
    (hbound : Integrable
      (fun x => ‖BiotSavart ω.val x‖ * ‖BiotSavart ε.val x‖ +
        ‖BiotSavart ε.val x‖ ^ 2)
      NavierStokes3D.volume)
    (hinner : AEStronglyMeasurable
      (fun x => inner ℝ (BiotSavart ω.val x) (BiotSavart ε.val x))
      NavierStokes3D.volume) :
    HasDerivAt
      (fun t : ℝ =>
        KineticEnergyHamiltonian
          (fun x => ω.val x + t • ε.val x,
            fun x => orbit_add_smul_div_free ω ε t hω hε x))
      (∫ x, pairing (velocity_from_vorticity ω x) (BiotSavart ε.val x)
        ∂volume) 0 := ...
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 224
public structure SDiffActionHyp (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
↪  :
    Prop where
  casimir_invariant : ∀ (f : ℝ → ℝ) (g : SDiff) (ω : CoadjointOrbit),
    (∫ x, f ‖(g • ω).val x‖ ∂volume) = ∫ x, f ‖ω.val x‖ ∂volume
```

```
-- FrohmanianTether/ConcreteSDiff.lean, line 98
theorem sdiffActionHyp : SDiffActionHyp G := ...
```

```
-- FrohmanianTether/ConcreteSDiff.lean, line 87
theorem casimir_invariant (f : ℝ → ℝ) (g : G) (ω : CoadjointOrbit) :
    (∫ x, f ‖(g • ω).val x‖ ∂volume) = ∫ x, f ‖ω.val x‖ ∂volume := ...
```

```
-- FrohmanianTether/Combined.lean, line 44
theorem left_H_zero_of_C2_symm
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F) (hC2 : DegenerateWRTKineticEnergy B)
```

```
    (ω : CoadjointOrbit) (M : Functional) :
    B ω KineticEnergyHamiltonian M = 0 := ...
```

```
-- FrohmanianTether/Combined.lean, line 52
theorem tetherKernel_left_H_zero (hC2 : DegenerateWRTKineticEnergy TetherKernel)
    (ω : CoadjointOrbit) (M : Functional) :
    TetherKernel ω KineticEnergyHamiltonian M = 0 ∧ TetherKernel ω M KineticEnergyHamiltonian
    ↪  = 0 := ...
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 1145
public theorem classicalBracket_antisymm (F G : CoadjointOrbit → ℝ) (ω : CoadjointOrbit) :
    ClassicalBracket F G ω = - ClassicalBracket G F ω := ...
```

```
-- FrohmanianTether/Uniqueness.lean, line 276
public def SaturatesTetherQuadratic
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (F : Functional) (ω : CoadjointOrbit), B ω F F = TetherKernel ω F F
```

```
-- FrohmanianTether/Uniqueness.lean, line 281
public def Polarizes
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ (ω : CoadjointOrbit) (F G : Functional),
    2 * B ω F G =
      B ω (fun ω' => F ω' + G ω') (fun ω' => F ω' + G ω') -
        B ω F F - B ω G G
```

```
-- FrohmanianTether/Uniqueness.lean, line 447
@[expose] public def FactorsThroughProjection
    (B : CoadjointOrbit → Functional → Functional → ℝ) : Prop :=
  ∀ ω : CoadjointOrbit, ∃ Q : (T3 → EuclideanSpace ℝ (Fin 3)) → (T3 → EuclideanSpace ℝ (Fin
  ↪  3)) → ℝ,
    (∀ a b, Q a b = Q b a) ∧
    ∀ F G : Functional, B ω F G =
      Q (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω))
        (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω))
```

```
-- FrohmanianTether/Uniqueness.lean, line 97
public def HasTetherKernelDensity
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (α : CoadjointOrbit → T3 → ℝ) : Prop :=
  ∀ (ω : CoadjointOrbit) (F G : Functional),
    B ω F G =
      ∫ x,
        α ω x *
          inner ℝ
            (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
            (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative G ω) x)
        ∂volume
```

```
-- FrohmanianTether/Uniqueness.lean, line 602
public theorem uniqueness_of_minimal_tether
    (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (_hyp : SDiffActionHyp SDiff)
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
```

```
    (hC1 : InvariantUnderCoadjointAction SDiff B)
    (hC2 : DegenerateWRTKineticEnergy B)
    (hC3 : ProducesControllableNegativeFeedback B)
    (h_sat : SaturatesTetherQuadratic B)
    (h_polarB : Polarizes B)
    (h_polarT : Polarizes TetherKernel) :
    ∀ (ω : CoadjointOrbit) (F G : Functional), B ω F G = TetherKernel ω F G := ...
```

```
-- FrohmanianTether/Uniqueness.lean, line 580
public theorem uniqueness_of_minimal_tether_core
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_sym : ∀ ω F G, B ω F G = B ω G F)
    (h_proj : ∀ ω (F F' G : Functional),
      Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) =
        Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F' ω) → B ω F G = B ω F' G)
    (h_density : FactorsThroughProjection B → ∃ α, HasTetherKernelDensity B α)
    (h_weight : ∀ α, HasTetherKernelDensity B α → ∃ c : ℝ, ∀ ω x, α ω x = -c * Enstrophy ω)
    (h_sat : SaturatesTetherQuadratic B)
    (h_nondeg : ∃ (ω : CoadjointOrbit) (F : Functional), Enstrophy ω * (∫ x,
        inner ℝ (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          (Pi_u (velocity_from_vorticity ω) (FunctionalDerivative F ω) x)
          ↪  ∂NavierStokes3D.volume) ≠ 0) :
    ∀ ω F G, B ω F G = TetherKernel ω F G := ...
```

```
-- FrohmanianTether/PlanSteps.lean, line 61
def IsCoadjointMotion (SDiff : Type*) [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (ω : ℝ → CoadjointOrbit) : Prop :=
  ∃ g : ℝ → SDiff, ∀ t, ω t = g t • ω 0
```

```
-- FrohmanianTether/PlanSteps.lean, line 67
theorem casimir_conserved {SDiff : Type*} [Group SDiff] [MulAction SDiff CoadjointOrbit]
    (H : SDiffActionHyp SDiff) (ω : ℝ → CoadjointOrbit) (hω : IsCoadjointMotion SDiff ω)
    (f : ℝ → ℝ) (t : ℝ) : Casimir f (ω t) = Casimir f (ω 0) := ...
```

```
-- FrohmanianTether/PlanSteps.lean, line 76
theorem tether_bound_not_scale_invariant (S C κ M : ℝ) (hκ : 0 < κ) (hM : M ≠ 0) :
    ¬ ∀ l : ℝ, 0 < l → l ^ 6 * S ≤ C * (l ^ 2 * M) ^ 3 - κ * (l ^ 2 * M) ^ 4 := ...
```

A.3 Refutations and degeneracy of the encoding

```
-- FrohmanianTether/ConcreteSDiff.lean, line 112
theorem not_transitive_on_casimir_levels :
    ¬ ∀ ω₁ ω₂ : CoadjointOrbit,
      (∀ f : ℝ → ℝ, Casimir f ω₁ = Casimir f ω₂) → ∃ g : G, g • ω₁ = ω₂ := ...
```

```
-- FrohmanianTether/StrainBoundRefutation.lean, line 78
public theorem not_forall_strain_le_sup_vorticity (C : ℝ) :
    ¬ ∀ (u : VelocityField) (x : T3),
      ‖fderiv ℝ u x‖ ≤ C * vorticity_sup_norm (curl u) := ...
```

```
-- FrohmanianTether/Uniqueness.lean, line 375
public theorem antisymmetric_clause_forces_zero
    (B : CoadjointOrbit → Functional → Functional → ℝ)
    (h_antisym : ∀ ω F G, B ω F G = -B ω G F)
    (h_sat : SaturatesTetherQuadratic B)
```

```
    (h_polarB : Polarizes B)
    (h_polarT : Polarizes TetherKernel) :
    ∀ (ω : CoadjointOrbit) (F G : Functional), B ω F G = 0 ∧ TetherKernel ω F G = 0 := ...
```

```
-- FrohmanianTether/ArnoldGeometric.lean, line 331
public theorem functionalDerivative_indep_of_not_differentiable
    {ω : CoadjointOrbit} (hω : ¬ ∀ i x, DifferentiableAt ℝ (fun y => ω.val y i) x)
    (F G : CoadjointOrbit → ℝ) :
    FunctionalDerivative F ω = FunctionalDerivative G ω := ...
```

```
-- FrohmanianTether/Uniqueness.lean, line 357
public theorem polarizes_tetherKernel_zero_of_not_differentiable
    (hP : Polarizes TetherKernel)
    {ω : CoadjointOrbit} (hω : ¬ ∀ i x, DifferentiableAt ℝ (fun y => ω.val y i) x)
    (F G : Functional) : TetherKernel ω F G = 0 := ...
```

A.4 Analytic layer

```
-- FrohmanianTether/AnalyticPipeline.lean, line 85
public theorem comparison_ode_stability
    (C κ'' : ℝ) (hC : 0 < C) (hκ : 0 < κ'')
    (y : ℝ → ℝ)
    (hdiff : ∀ s ≥ (0 : ℝ), DifferentiableAt ℝ y s)
    (hy_dot : ∀ s ≥ (0 : ℝ), deriv y s ≤ C * (y s) ^ 2 - κ'' * (y s) ^ 3)
    (t : ℝ) (ht : 0 ≤ t) :
    y t ≤ max (y 0) (C / κ'') := ...
```

```
-- FrohmanianTether/TetheredLyapunov.lean, line 2150
public theorem riccati_ceiling_of_vorticity_di
    (u : TimeDependentVelocity) (C κ'' : ℝ)
    (hC : 0 < C) (hκ : 0 < κ'')
    (hdiff : ∀ s ≥ (0 : ℝ),
      DifferentiableAt ℝ (fun τ => vorticity_sup_norm (vorticity (u τ))) s)
    (hDI : ∀ s ≥ (0 : ℝ),
      deriv (fun τ => vorticity_sup_norm (vorticity (u τ))) s ≤
        C * vorticity_sup_norm (vorticity (u s)) ^ 2 -
          κ'' * vorticity_sup_norm (vorticity (u s)) ^ 3)
    (hcont : Continuous fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) :
    ∃ Y : ℝ,
      (∀ τ ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u τ)) ≤ Y) ∧
      Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) := ...
```

```
-- FrohmanianTether/SymplecticTether.lean, line 556
@[expose] public abbrev FrohmanianTetherRate (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    0 < ν → NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u →
      ∃ C κ'' : ℝ, 0 < C ∧ 0 < κ'' ∧
        (∀ s ≥ (0 : ℝ),
          DifferentiableAt ℝ (fun τ => vorticity_sup_norm (vorticity (u τ))) s) ∧
        (∀ s ≥ (0 : ℝ),
          deriv (fun τ => vorticity_sup_norm (vorticity (u τ))) s ≤
            C * vorticity_sup_norm (vorticity (u s)) ^ 2 -
              κ'' * vorticity_sup_norm (vorticity (u s)) ^ 3) ∧
        Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) ∧
        (∀ τ ≥ (0 : ℝ), BddAbove (Set.range fun x => ‖vorticity (u τ) x‖))
```

```
-- FrohmanianTether/SymplecticTether.lean, line 544
@[expose] public abbrev FrohmanianTetherBound (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u →
      ∃ Y : ℝ,
        (∀ τ ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u τ)) ≤ Y) ∧
        Continuous (fun τ : ℝ => vorticity_sup_norm (vorticity (u τ))) ∧
        (∀ τ ≥ (0 : ℝ), BddAbove (Set.range fun x => ‖vorticity (u τ) x‖))
```

```
-- FrohmanianTether/ClayStatements.lean, line 47
@[expose] public def FrohmanianTetherClosingBound (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) →
        FrohmanianTetherBound u₀ ν
```

```
-- FrohmanianTether/NS_Equations.lean, line 3086
@[expose] public def BKMContinuationCriterion (u : TimeDependentVelocity) (ν : ℝ) : Prop :=
  ∀ T : ℝ, Tstar (u 0) ν = (T : EReal) →
    ¬ IntegrableOn (fun t => vorticity_sup_norm (vorticity (u t))) (Set.Icc 0 T)
```

```
-- FrohmanianTether/TetheredLyapunov.lean, line 1843
@[expose] public def PhysicalBKMContinuationCriterionFor (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∀ u : ℝ → VelocityField, ∀ p : ℝ → PressureField,
    NS_PDE u p ν → u 0 = u₀ → IsPhysicalNSSolution u → BKMContinuationCriterion u ν
```

```
-- FrohmanianTether/TetheredLyapunov.lean, line 1857
@[expose] public def KatoUniquenessOnExistenceTimes (u₀ : VelocityField) (ν : ℝ)
    (w : KatoLocalWitness u₀ ν) : Prop :=
  ∀ (T' : ℝ) (v : TimeDependentVelocity) (q : TimeDependentPressure),
    T' ∈ existenceTimes u₀ ν → v 0 = u₀ → NS_PDE v q ν →
    (∀ t ∈ Set.Icc (0 : ℝ) T', ContDiff ℝ (⊤ : ℕ∞) (v t)) →
    IsPhysicalNSSolution v →
    ∀ t ∈ Set.Icc (0 : ℝ) T', w.u t = v t
```

```
-- FrohmanianTether/ClosureBlueprint.lean, line 215
public theorem conditional_regularity_of_open_hyps
    (u₀ : VelocityField) (ν : ℝ)
    (h_divfree : ∀ x, div u₀ x = 0)
    (h_smooth : ContDiff ℝ (⊤ : ℕ∞) u₀)
    (h_finite_energy : Integrable (fun x : T3 => ‖u₀ x‖ ^ 2))
    (h_pos_ν : ν > 0)
    (w : KatoLocalWitness u₀ ν)
    (hRiccati : FrohmanianTetherBound u₀ ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor u₀ ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes u₀ ν w) :
    ∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
      NS_PDE u p ν ∧
      u 0 = u₀ ∧
      (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
      (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0) := ...
```

```
-- FrohmanianTether/ConditionalRegularity.lean, line 165
public theorem conditional_regularity_of_open_hyps
```

```
    (u0 : TimeDependentVelocity)
    (h_div_free : ∀ t x, div (u0 t) x = 0)
    (h_smooth : IsSmooth (u0 0))
    (h_finite_energy : Integrable (fun x : T3 => ‖u0 0 x‖ ^ 2))
    (ν : ℝ) (h_ν_pos : ν > 0)
    (w : KatoLocalWitness (u0 0) ν)
    (hRiccati : FrohmanianTetherBound (u0 0) ν)
    (hcrit : TetheredLyapunov.PhysicalBKMContinuationCriterionFor (u0 0) ν)
    (huniq : TetheredLyapunov.KatoUniquenessOnExistenceTimes (u0 0) ν w) :
  ∃ (u : TimeDependentVelocity),
    (∀ t ≥ (0 : ℝ), IsSmooth (u t)) ∧
    (u 0 = u0 0) ∧
    (∀ t ≥ (0 : ℝ), satisfies_NavierStokes u ν) ∧
    (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
    (∀ v : TimeDependentVelocity,
      (∀ t ≥ (0 : ℝ), IsSmooth (v t)) →
      v 0 = u0 0 →
      (∀ t ≥ (0 : ℝ), satisfies_NavierStokes v ν) →
      ∀ t ≥ (0 : ℝ), v t = u t) := ...
```

```
-- FrohmanianTether/ConditionalRegularity.lean, line 93
public theorem frohmanian_tether_theorem
    (hKato : ∀ u₀ ν, ContDiff ℝ (⊤ : ℕ∞) u₀ → (∀ x, div u₀ x = 0) →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) → 0 < ν →
      KatoLocalWitness u₀ ν)
    (hDI : ∀ u₀ ν, FrohmanianTetherRate u₀ ν)
    (hcrit : ∀ u₀ ν, TetheredLyapunov.PhysicalBKMContinuationCriterionFor u₀ ν)
    (huniq : ∀ u₀ ν (w : KatoLocalWitness u₀ ν),
      TetheredLyapunov.KatoUniquenessOnExistenceTimes u₀ ν w) :
  ∃ (𝔗_F : CoadjointOrbit → Functional → Functional → ℝ),
    (∀ (B : CoadjointOrbit → Functional → Functional → ℝ)
      (α : CoadjointOrbit → T3 → ℝ),
      HasTetherKernelDensity B α →
      (∀ ω x, α ω x = canonicalTetherDensity ω x) →
      ∀ ω F G, B ω F G = 𝔗_F ω F G) ∧
    (∀ (u₀ : VelocityField) (ν : ℝ),
      0 < ν →
      (∀ x, div u₀ x = 0) →
      ContDiff ℝ (⊤ : ℕ∞) u₀ →
      Integrable (fun x : T3 => ‖u₀ x‖ ^ 2) →
      ∃ (u : ℝ → VelocityField) (p : ℝ → PressureField),
        NS_PDE u p ν ∧
        u 0 = u₀ ∧
        (∀ t ≥ (0 : ℝ), ContDiff ℝ (⊤ : ℕ∞) (u t)) ∧
        (∀ t ≥ (0 : ℝ), vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
        (∀ t ≥ (0 : ℝ), ∀ x, div (u t) x = 0)) := ...
```

```
-- FrohmanianTether/ClayStatements.lean, line 477
public theorem conditional_regularity_nonvacuous_at_zero (ν : ℝ) (hν : 0 < ν) :
    ∃ (u : TimeDependentVelocity),
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (u t)) ∧
      (u 0 = 0) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes u ν) ∧
      (∀ t ≥ (0 : ℝ), ConditionalRegularity.vorticity_sup_norm (vorticity (u t)) ≥ 0) ∧
      (∀ v : TimeDependentVelocity,
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.IsSmooth (v t)) →
        v 0 = 0 →
        (∀ t ≥ (0 : ℝ), ConditionalRegularity.satisfies_NavierStokes v ν) →
        ∀ t ≥ (0 : ℝ), v t = u t) := ...
```

```
-- FrohmanianTether/ClayStatements.lean, line 396
```

```
public theorem stepField_not_physical : ¬ IsPhysicalNSSolution stepField := ...
```

A.5 Periodic Clay statement

```
-- FrohmanianTether/ClayStatements.lean, line 250
@[expose] public def Clay_B_exact : Prop :=
  ∀ (ν : ℝ) (u₀ : VelocityField),
    0 < ν →
    ContDiff ℝ (⊤ : ℕ∞) u₀ →
    (∀ x, div u₀ x = 0) →
    IsZ3Periodic u₀ →
    ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
      IsPeriodicSmoothSolution ν u₀ (fun _ _ => 0) u p
```

```
-- FrohmanianTether/ClayStatements.lean, line 528
@[expose] public def PeriodicTetherClosingBound (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ → IsZ3Periodic u₀ →
      ∃ Y : ℝ, ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
        IsPeriodicSmoothSolutionOn ν T u₀ u p →
          ∀ t ∈ Set.Ico (0 : ℝ) T, vorticity_sup_norm (vorticity (u t)) ≤ Y
```

```
-- FrohmanianTether/ClayStatements.lean, line 540
@[expose] public def PeriodicBKMContinuation (ν : ℝ) : Prop :=
  ∀ u₀ : VelocityField,
    (∀ x, div u₀ x = 0) → ContDiff ℝ (⊤ : ℕ∞) u₀ → IsZ3Periodic u₀ →
      (∃ Y : ℝ, ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
        IsPeriodicSmoothSolutionOn ν T u₀ u p →
          ∀ t ∈ Set.Ico (0 : ℝ) T, vorticity_sup_norm (vorticity (u t)) ≤ Y) →
        ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
          IsPeriodicSmoothSolution ν u₀ (fun _ _ => 0) u p
```

```
-- FrohmanianTether/ClayStatements.lean, line 552
public theorem clay_B_exact_of_periodic_tether_bound
    (hclose : ∀ ν > (0 : ℝ), PeriodicTetherClosingBound ν)
    (hBKM : ∀ ν > (0 : ℝ), PeriodicBKMContinuation ν) : Clay_B_exact := ...
```

B Verification certificate

Repository	https://github.com/BenFrohman/frohmanian-tether-sdiff, branch `remove-kinetic-energy-hypothesis` (version 0.1.2 draft, unreleased)
Lean toolchain	`leanprover/lean4:v4.30.0-rc1`
Mathlib	`v4.30.0-rc1`, rev `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
Build	`lake build`: 2715 jobs, success
Modules	25 (`FrohmanianTether` and 24 submodules)
Theorems audited	507 (466 stated in the sources, 41 auto-generated)
Declarations audited	750 (theorems, definitions, structures, instances)
Axioms	only `propext`, `Classical.choice`, `Quot.sound`
`sorry` / `admit` / `native_decide`	none
Axiom declarations	none

The exact commit, the CI runs, and the full per-theorem axiom list are recorded in `CERTIFICATE.md` in the repository. Reproduction:

```
lake exe cache get
lake build
python3 scripts/check_no_sorry.py
lake env lean scripts/AxiomAudit.lean
```

`scripts/AxiomAudit.lean` computes `collectAxioms` for every non-internal declaration in the `FrohmanianTether` modules and raises an error (non-zero exit) if any axiom other than the three standard ones occurs (this includes `sorryAx`, produced by `sorry`, and `Lean.ofReduceBool`, produced by `native_decide`) or if any axiom is declared. `scripts/check_no_sorry.py` rejects the tokens `sorry`, `admit`, `native_decide` and `axiom` declarations outside comments and strings. Output for the headline declarations:

```
'ArnoldGeometric.functionalDerivative_indep_of_not_differentiable' depends on axioms:
↪  [propext, Classical.choice, Quot.sound]
'Combined.left_H_zero_of_C2_symm' depends on axioms: [propext, Classical.choice, Quot.sound]
'Combined.tetherKernel_left_H_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'ConcreteSDiff.not_transitive_on_casimir_levels' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
'ConcreteSDiff.sdiffActionHyp' depends on axioms: [propext, Classical.choice, Quot.sound]
'ConditionalRegularity.frohmanian_tether_theorem' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
'FrohmanianTether.antisymmetric_clause_forces_zero' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
'FrohmanianTether.polarizes_tetherKernel_zero_of_not_differentiable' depends on axioms:
↪  [propext, Classical.choice, Quot.sound]
'KineticEnergyDerivative.not_forall_functionalDerivative_kineticEnergy_eq_velocity' depends on
↪  axioms: [propext, Classical.choice, Quot.sound]
'PhysicalSolutionSanity.exists_physical_time_deriv_ne_zero' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
'StrainBoundRefutation.not_forall_strain_le_sup_vorticity' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
AUDIT SUMMARY: theorems audited: 507 (466 stated in the sources, 41 auto-generated
  equation/congruence/injectivity lemmas); declarations audited: 750; violations: 0
AUDIT PASS: only propext, Classical.choice, Quot.sound; no sorryAx; no axiom declarations
```

Licence

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